Meric Altug Gemalmaz, Ph.D.

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RESEARCH INTERESTS

Human-AI Interaction, Crowdsourcing and Human Computation, Applied AI, Fairness

EDUCATION

Purdue University, West Lafayette, Indiana

- Ph.D. in Computer Science **3.96/4.0** Jan. 2020 – June 2025
- M.S. in Computer Science (Transferred to Ph.D.) **3.93/4.0** Jan. 2019 – Dec. 2019
- B.S. in Computer Science (Software Engineering Track, **Distinction**) **3.92/4.0** Aug. 2015 – Dec. 2018

SKILLS

Programming Languages: Python, C/C++, C#, JavaScript, Java, HTML/CSS, R, SQL, Bash
Machine Learning Tools: scikit-learn, GSL, NumPy, Pandas
Tools & Platforms: Git, Meteor.js, MongoDB, MTurk, Android SDK, Firebase, GDB

PUBLICATIONS

[P1] Meric Altug Gemalmaz, Ming Yin. "Understanding Decision Subjects' Fairness Perceptions and Retention in Repeated Interactions with AI-Based Decision Systems." *Proceedings of the 5th AAAI/ACM Conference on AI, Ethics, and Society (AIES)*, Oxford, UK, Aug. 2022.

[P2] Meric Altug Gemalmaz, Ming Yin. "Accounting for Confirmation Bias in Crowdsourced Label Aggregation." *Proceedings of the 30th International Joint Conference on Artificial Intelligence (IJCAI)*, Online, Aug. 2021.

[P3] Meric Altug Gemalmaz. "Fairness in the Machine Learning Pipeline: A Human-in-the-Loop Perspective." *Ph.D. Dissertation, Department of Computer Science, Purdue University, West Lafayette, Indiana, USA*, June 2025.

WORK INCLUDED IN PH.D. DISSERTATION

[W1] Meric Altug Gemalmaz, Ming Yin. "An Investigation of Decision Subjects' Interaction With Periodically Updated ML-Based Task Allocations." *(See Chapter 5)*

[W2] Meric Altug Gemalmaz, Ming Yin. "Understanding Decision Subjects' Engagement with and Perceived Fairness of AI Models When Opportunities of Qualification Improvement Exist." *arXiv:2410.03126 (See Chapter 4)*

INDUSTRY EXPERIENCE

ML Research Scientist, Meta, Menlo Park, CA Aug. 2025 – Present

- Contributed to large-scale recommendation and ranking systems, first on content recommendation and content quality for Facebook Groups, now on data and evaluation for AI agent systems.
- Shipped **17** A/B-tested launches in roughly nine months on Groups, across **two** distinct recommendation surfaces and tech stacks: built LLM-based content-quality signals, **two** new candidate generators, and a set of content-integrity detection signals.
- Upheld engineering excellence by ramping onto two tech stacks in my first month, taking production oncall by month four, and always chasing root causes in retrieval and measurement instead of patching symptoms.
- Improve the data AI models learn from for AI agent systems, by building the workflows that create and improve their data, alongside creating evaluations for their decisions.

RESEARCH EXPERIENCE

Human-subject Experiments Jan. 2020 – May 2025

- Coordinated **1,500+** human participants across **4+** research projects on **MTurk** for data collection and analysis.
- Developed and deployed web applications using **JavaScript** and the **Meteor.js** framework, managed back-end databases with **MongoDB**, and implemented front-end interfaces using **HTML** and **CSS**.
- Delivered 90% participant satisfaction and blocked 80% of the bot attacks the system received by designing scenario-based, interactive UIs.

Exploring Fairness in Algorithmic Management [W1] Dec. 2023 – May 2025

- Collaborated with gig workers to explore their long-term behavior toward AI-driven gig assignments with varying levels of fairness, yielding actionable insights for fairer algorithmic job management.

Understanding Decision Subjects' Fairness Perceptions and Engagement [W2, P1] Mar. 2021 – Dec. 2023

- Conducted six human-subject experiments to examine how loan applicants' repeated interactions with an AI-based loan approval system affect their fairness perceptions and willingness to continue engaging with the AI.
- Simulated loan applicants' AI interactions with a **Markov Decision Process** to estimate proper human-subject experiment parameters, then collected data to analyze real human-AI interactions with **regression analysis** to understand human behavior.

- Discovered a critical fairness issue: similar AI usage across demographics often hides unfairness, as people continue using biased models out of necessity, not fairness. This persistence challenges developers to rethink usage as a measure of model fairness.

Data Bias Mitigation Algorithm [P2] Mar. 2020 – Mar. 2021

- Leveraged advanced unsupervised learning techniques to detect and mitigate cognitive bias in crowdsourced data annotations.
- Utilized **probabilistic graphical models** to capture annotator bias and used the **Expectation-Maximization algorithm** to infer ground-truth annotations.
- Achieved over **10% increase** in inferred label accuracy over existing baselines through reduction in annotation bias.

SELECT COURSES

Graduate Courses: Data Mining (A), Algorithm Design, Analysis, And Implementation (A+), Statistical Machine Learning (A-), Simulation And Modeling Of Computer Systems (A), Operating Systems (A)

Undergraduate Courses: Software Engr Senior Project (A), Software Engineering I (A), Object-Oriented Programming (A), Programming In C (A+), Foundations Of Computer Science (A-)

TEACHING EXPERIENCE

Graduate Teaching Assistant, Purdue University Jan. 2019 – May 2025

- **Courses:** Data Mining (2 semesters), Systems Programming (4 semesters), Computer Architecture (3 semesters).
- Led lab sections each semester for **50+** students, supervised **10+** undergraduate TAs, collaborated with GTAs to develop teaching materials, and managed administrative responsibilities.
- Received excellent teaching evaluations (avg. **4.5**/**5.0**) and awarded multiple teaching awards for leadership, communication, and adaptability.
- **Guest lecturer** on Human-Computer Interaction; Lecture on AI Ethics and Fairness
- **Guest lecturer** on Systems Programming; Lecture on Implementation of a Concurrent Web Server

Undergraduate Teaching Assistant, Purdue University Jan. 2018 – Dec. 2018

- **Courses:** Systems Programming (2 semesters), Operating Systems (1 semester).
- Supported lab sessions, provided individualized student support, and collaborated on grading assignments.
- Developed scripts in **Bash** to streamline the evaluation process.

HONORS AND AWARDS

The Graduate Teaching Award Nov. 2022

- Recognized for exceptional teaching and leadership in graduate level Data Mining course. Awarded based on positive feedback from faculty and students, highlighting approachability, support, and ability to clarify complex concepts.

Raymond Boyce Graduate Teaching Award Apr. 2020

- Honored for outstanding contributions to the Computer Architecture course. Received recognition on a permanent plaque in the Lawson Computer Science Building.

Graduated with Distinction, B.S. in Computer Science Dec. 2018

- Awarded to the top **10%** of undergraduate Computer Science students for exceptional academic performance.

Dean's List and Semester Honors (8 Semesters) Aug. 2015 – Dec. 2018

- Placed consistently on the Dean's List and received Semester Honors during all semesters of the CS undergraduate program.

REVIEWER

Conference Reviewer; *Outstanding Reviewer Recognition
ACM CHI: 2024*

Program Committee Member
ACM IUI: 2025

REFERENCES

Prof. Ming Yin
Computer Science, Purdue University
Areas: Human-AI Interaction, Crowdsourcing
mingyin@purdue.edu

Prof. Sooyeon Jeong
Computer Science, Purdue University
Areas: Human-Robot Interaction, Social Robotics
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Prof. Dan Goldwasser
Computer Science, Purdue University
Areas: NLP, Large Language Models, Machine Learning
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Prof. Chris Clifton
Computer Science, Purdue University
Areas: Data Privacy, Data Mining, Database Systems
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